Introduction to Trade Studies, Risk, and Decision Analysis

Trade Studies and Decision Analysis in Systems Engineering

Typical Decisions

- Problems that may require a formal decision process.
  - Trade studies (trade off studies; trade-off studies)
  - Bid/no-bid
  - Make-reuse-buy
  - Formal inspection versus checklist inspection
  - Tool and vendor selection
  - Cost estimating
  - Architectural design
  - Hiring and promotions
  - Helping customer choose a solution

Trade Studies

- A tradeoff study is not something that is done once at the beginning of a project.
- Throughout a project there are several tradeoff studies performed.
  - Creating team communication methods
  - Selecting components
  - Choosing implementation techniques
  - Designing test programs
  - Maintaining schedule
- Many of these trade studies need be formally documented.

Trade Studies

- Refer to both different ends and different methods.
- All proposed alternatives are not necessarily attainable.
  - Better to consider many alternatives than to overlook one that might be preferred.
  - Alternatives that are not considered cannot be adopted.
- Alternative should consist of a complete description of its objectives and its requirements in terms of both benefit and cost.

Trade Studies

- In addition to identified alternatives, there is always another alternative: no decision.
  - The decision not to decide is motivated by the thought that there will be opportunities in the future that may prove more desirable than any known at present.
- Limiting factors.
  - Factors restricting accomplishment of desired objectives.
- Strategic factors.
  - Factors that can be altered to make progress possible.

Trade Studies

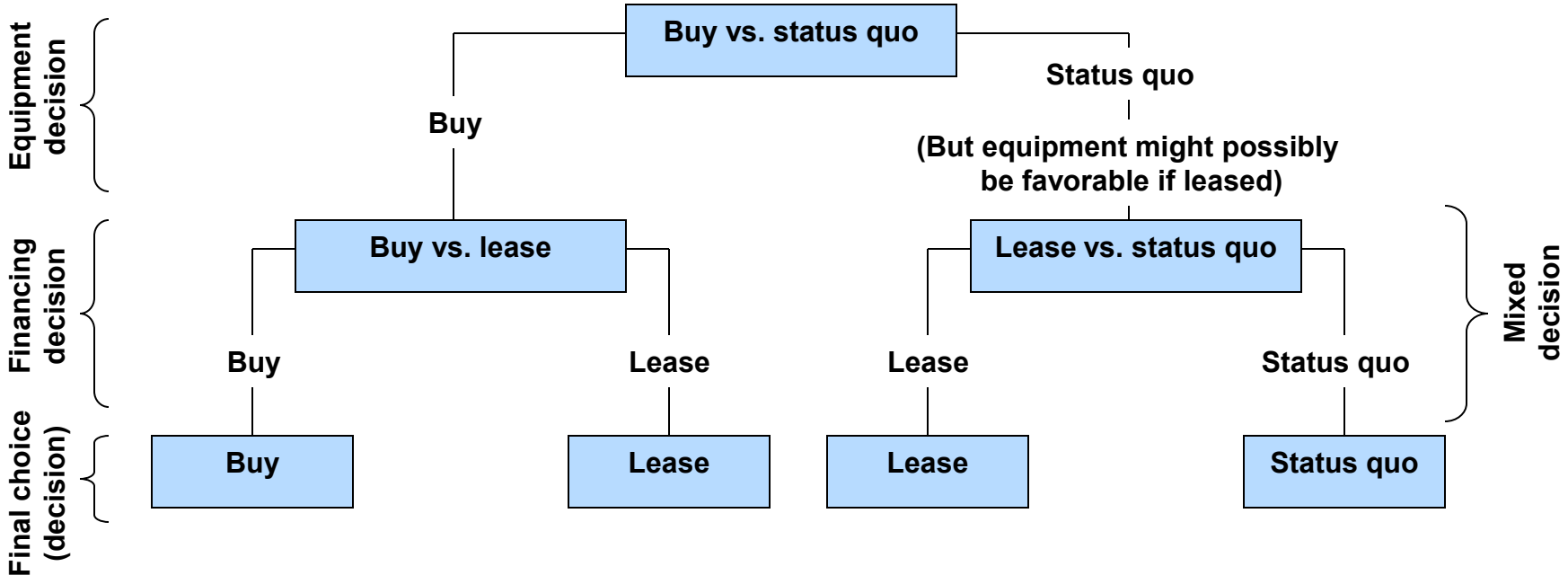
- Comparing alternatives equivalently.
  - Need to be converted to a common measure.
  - Use money flow models, economic optimization models.
- Decision made with the involvement of the customer.
- Several outcomes are evaluated for each alternative examined.
  - All outcomes cannot occur.
  - At any given time, any outcome can occur.
- Decision is made often under risk and uncertainty.

Real-life Trade Studies

- When to use?
  - Buying a car, buying a house, selecting a job.
  - These decisions are important, you have lots of time to make the decision and alternatives are apparent.
- When not to use?
  - Select a drink for lunch.
- Also, you would do a tradeoff study when your boss asks you to do one.

Real-life Trade Studies

- In the case of leasing decisions, the firm needs to first consider the buy versus status quo decision, and then consider either the buy versus lease decision (if buying is preferable to status quo) or the lease versus status quo decision (if status quo is preferable to buying).



Trade Studies Models

- Representation of a system to be brought into being, or to analyze an existing system.
  - Decisions made in less time and at a less cost than direct manipulation of the system.
- Direct and indirect experimentation.
  - Simulation.
- Formulating and validating models.
  - Include factors on which the outcome is believed to depend significantly.
  - Manipulate model to reduce the misfit between the model and the real-world.
    - Postulate a model.
    - Test model prediction against observations.
    - Modify the model to reduce misfit.

Trade Studies Models

- Physical models.
  - Geometric equivalents, either as miniatures, enlargements, or duplicates made to the same scale.
  - Globe, model of atomic structure, wind tunnel studies.
- Analogue models.
  - Build for similarity in relations.
  - Electric circuits used to represent mechanical systems, hydraulic systems.
- Schematic models.
  - Developed by reducing a state or event to a chart or diagram.
  - Organizational chart, process flow chart.

Trade Studies Models

- Mathematical models.
  - Description of the system using the language of mathematics used to predict or control.
  - Newton’s law, Ohm’s law.

Decisions…

- Humans are not always rational.
  - Emotions, cognitive illusions, biases, fear of regret make humans far from ideal decision makers.
  - Most people would be more concerned with a large potential loss than with a large potential gain. Losses are felt more strongly than equal gains
- Using trade studies judiciously can help make rational decisions.

Decisions

- To a large extent, the successes or failures that a person experience in life depend on the decisions he or she makes.
  - The person who managed the ill-fated space shuttle Challenger is no longer working for NASA.
  - The person who designed the top-selling Mustang became president at Ford.
- What makes the difference between good and bad decisions?

Good Decisions… Bad Decisions...

- "Sir, What is the secret of your success?" a reporter asked a bank president.
  "Two words."
  "And, sir, what are they?"
  "Good decisions."
  "And how do you make good decisions?"
  "One word."
  "And sir, what is that?"
  "Experience."
  "And how do you get Experience?"
  "Two words."
  "And, sir, what are they?"
  "Bad decisions."

  - Unknown author

Good Decisions

- Are based on logic.
- Consider all available data and possible alternatives.
- Employs formal quantitative approach.

- A good decision may, occasionally, result in an unexpected or unfavorable outcome; however, it is still a good decision if it is made properly.

Bad Decisions

- Are not based on logic.
- Do not use all available information and do not consider all possible alternatives.
- Do not employ appropriate quantitative techniques.

- If one makes a bad decision, but he/she is lucky and a favorable outcome occurs, he/she still made a bad decision.

Risk

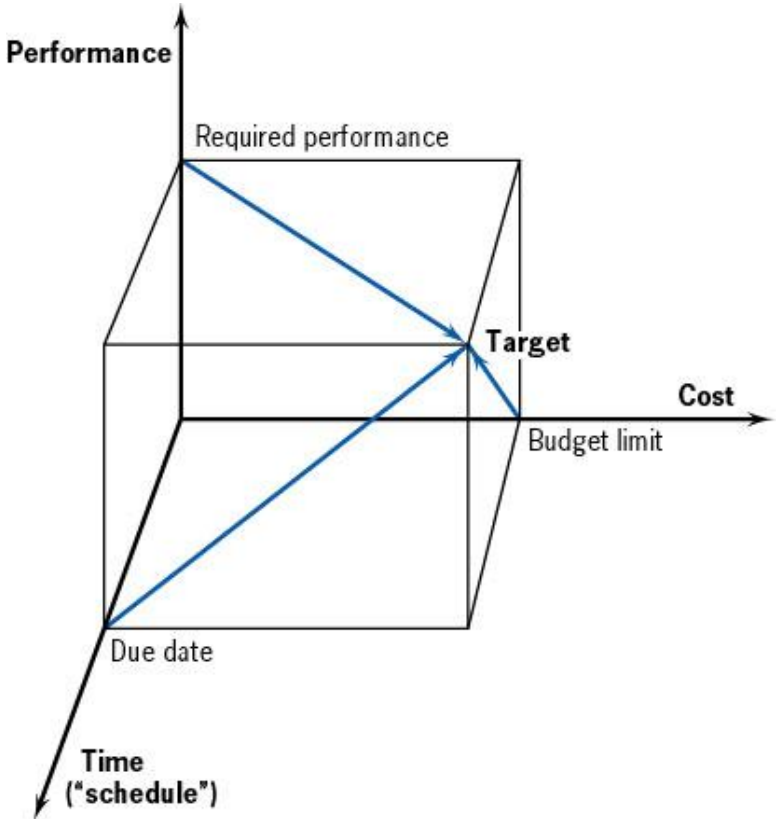
- Possibility of loss or injury.
- An event that, if it occurs, has unwanted consequences.
- Engineering: "What can go wrong with my system or any of its parts?" (Kaplan and Garrick, 1981).
- Probability theory: "The chance an unwanted event occurs" (Hansson, 2008).
- Economics: the way a person evaluates the monetary worth of participation in a lottery or a gamble.
  - Risk-averse person: if willing to accept with certainty an amount of money less than the expected amount potentially received from lottery.

Risk in Engineering

- Events that, if they occur, cause unwanted change in the cost, schedule, or technical performance of an engineering system.
- Event that may occur with probability p or may not occur with probability $(1 - p)$.
- Why are risks present?
  - Pressure to meet cost, schedule, and technical performance.
  - Expectations in these dimensions push what is technically or economically feasible.
  - Risk management is managing the inherent contention that exists within and across these dimensions.

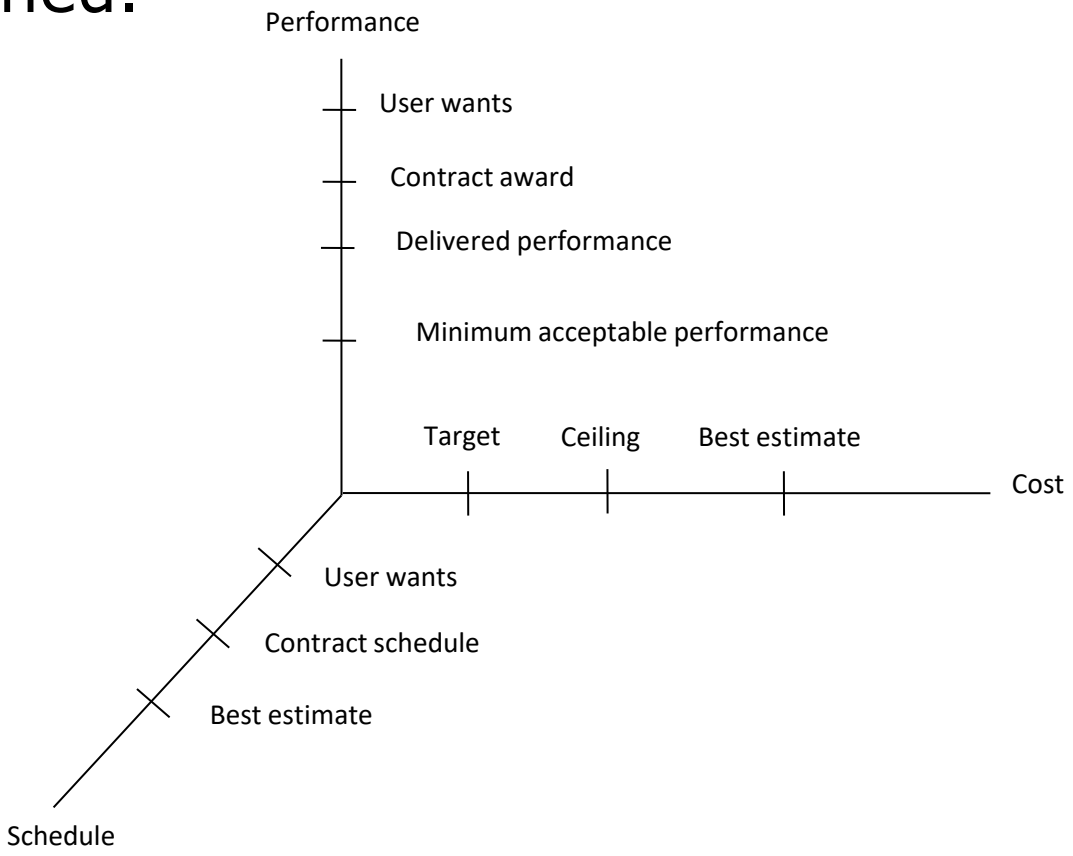
Risk in Engineering

- Projects are evaluated using three criteria.
  - Time
    - Agreed by the client
  - Cost
    - Agreed by the client
  - Performance
    - Set by the client
    - Capabilities required



Risk in Engineering

- Driving consideration in decisions that determine how engineering systems are developed, produced, and sustained.

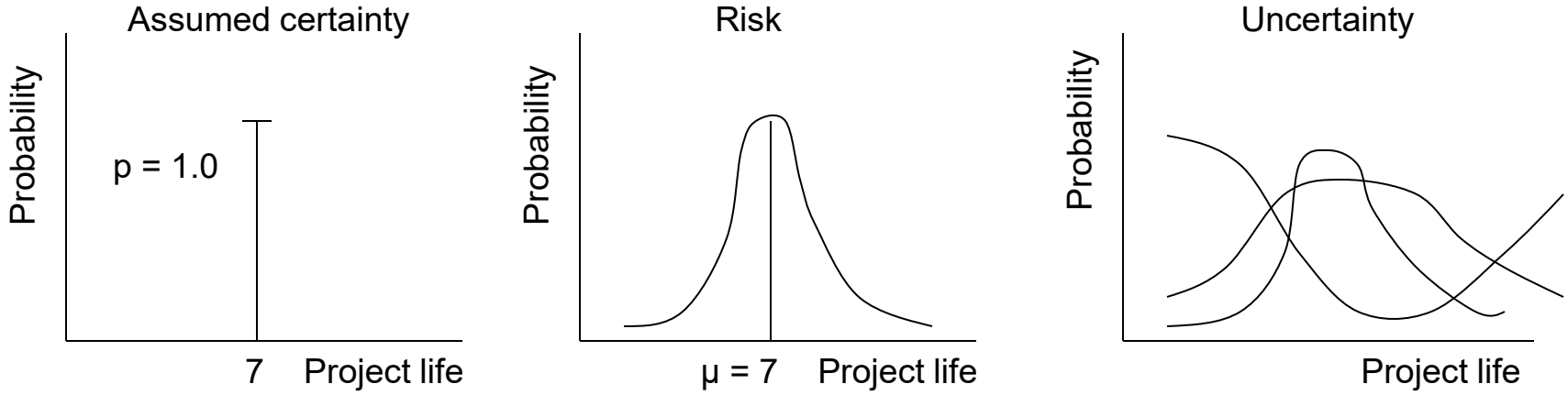


Certainty, Risk and Uncertainty

- Certainty
  - Possible outcomes are known with certainty.
- Risk
  - Probabilities of the possible outcomes are known.
- Uncertainty
  - Frequency distribution of the possible outcomes is not known.

Certainty, Risk and Uncertainty

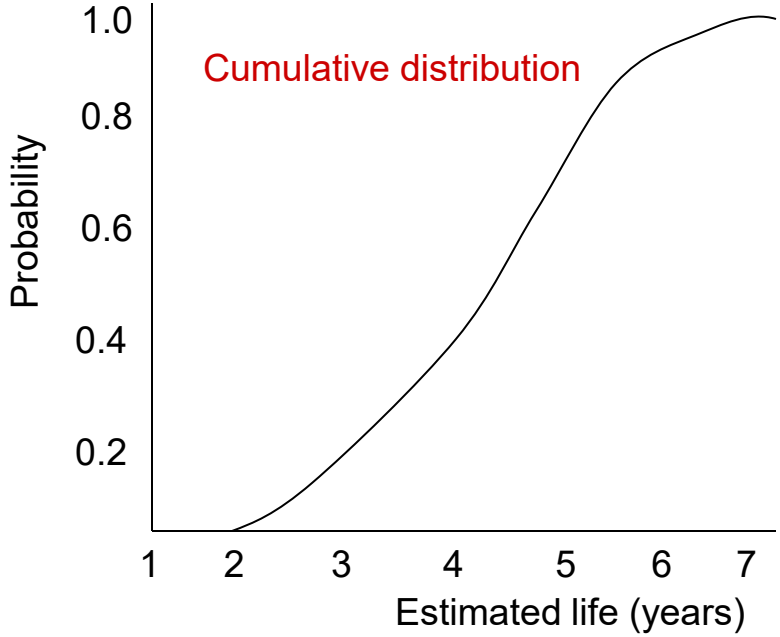
- Certainty
  - The outcome occurs with an assumed certainty.
- Risk
  - Dispersion of the probability distribution of the outcome.
- Uncertainty
  - Degree of lack of confidence that the estimated probability distribution of the outcome is correct.

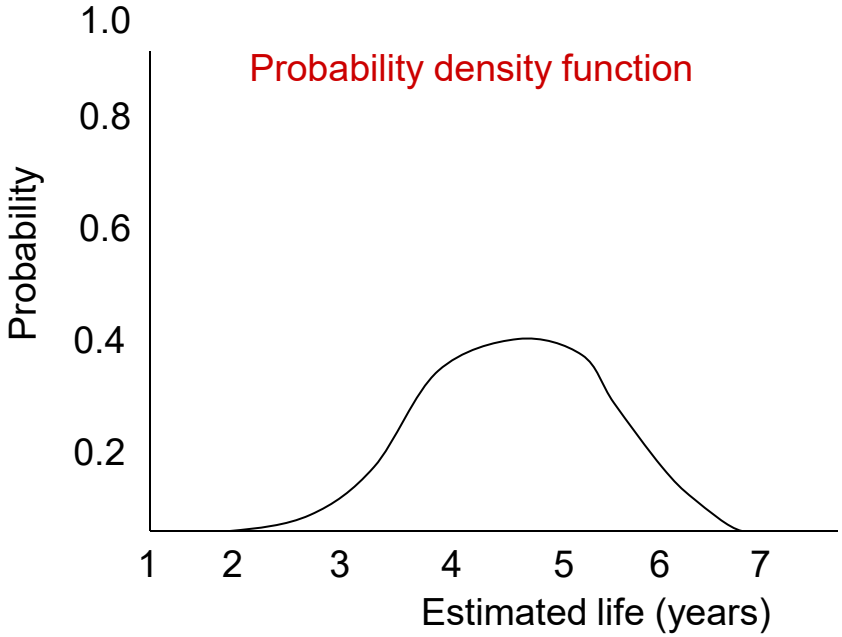


Estimating Risk in Terms of Probability Distributions

Life (years)	Probability life is less than or equal to the estimated life
2	0.0
3	0.1
4	0.3
5	0.7
6	0.9
7	1.0

Life (years)	Probability of life
2	0.0
3	0.1
4	0.3
5	0.4
6	0.2
7	0.0



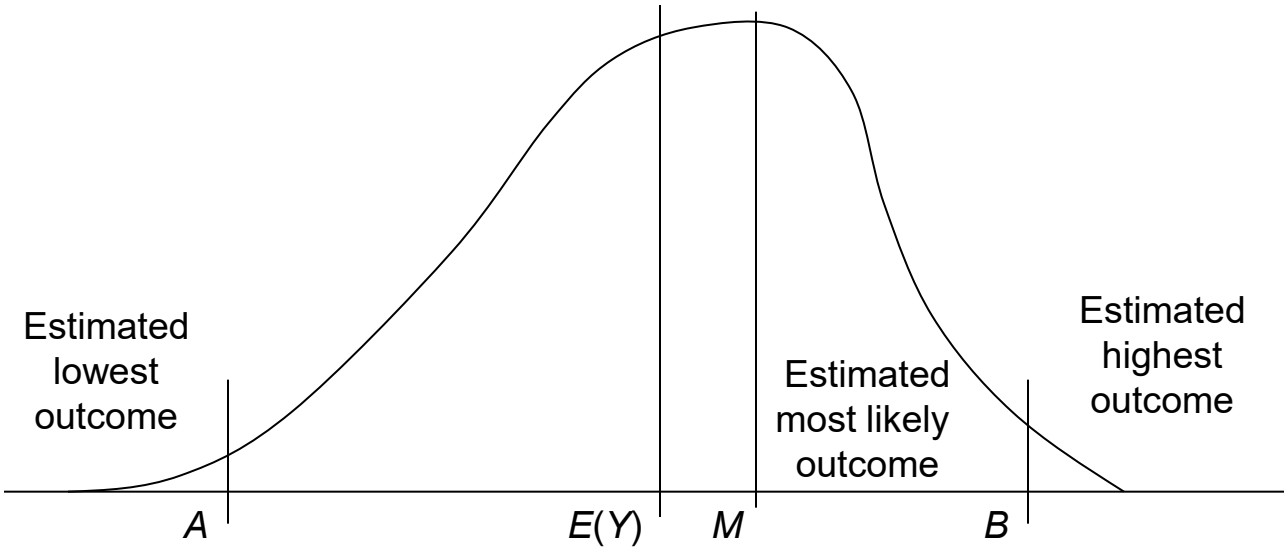


Probability that life is less than or equal to the estimated life

Probability of life

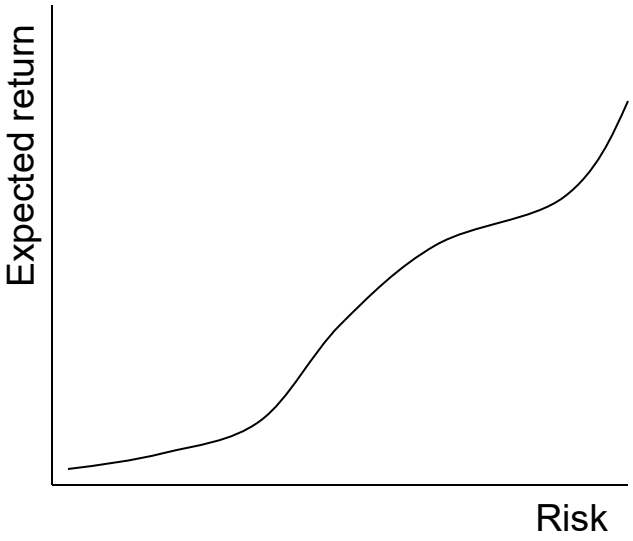
Estimating Risk in Terms of Probability Distributions

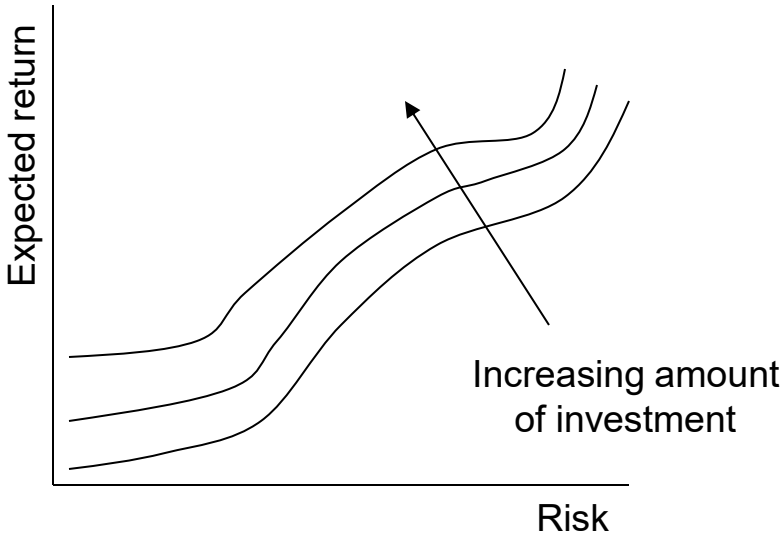
- Beta distribution
  - Optimistic: A – estimated lowest outcome
  - Most likely: M – estimated most likely outcome (mode)
  - Pessimistic: B – estimated highest outcome
  - $E(Y) = (A+4M+B)/6$ – estimated expected outcome
  - $V(Y) = [(B-A)/6]^2$ – estimated variance of outcome



Risk and Return

- Project acceptance under risk.
- The higher the assessed risk, the higher should be the expected return.





Causes of Risk and Uncertainty

- Insufficient number of similar investments
  - A firm invests in a few projects of the same type.
  - Insufficient opportunity for the results to "average out" (effect of unfavorable outcomes to be cancelled by favorable outcomes).
  - Dominant when investments are large.
- Bias in the data and its assessment
  - Optimism or pessimism of the part of the analyst.
  - Patterns should be recognized through review procedures.
- Changing external economic environment, invalidating past experience
  - Using past information without adjusting it for future conditions.
- Misinterpretation of data
  - Underlying factors and relationships are misunderstood.

Risk Analysis

- Increases probability of project success.
  - Potential to reduce project schedule and cost.
- Facilitates proper allocation of resources.
  - Improves project planning.
- Communicates project status.
  - Management and customers are always interested in risk.

Flight Risk

- Let's say the probability of a jet engine's failure, for any reason during a commercial flight is 1 in 100,000.
  - Acceptable risk?
- Fortunately, each jet has at least two engines.
- Safety engineer calculates that the probability of both engines' failure is $(1/100{,}000)^2$.
  - Reasonable?

Real-life Risk Analysis

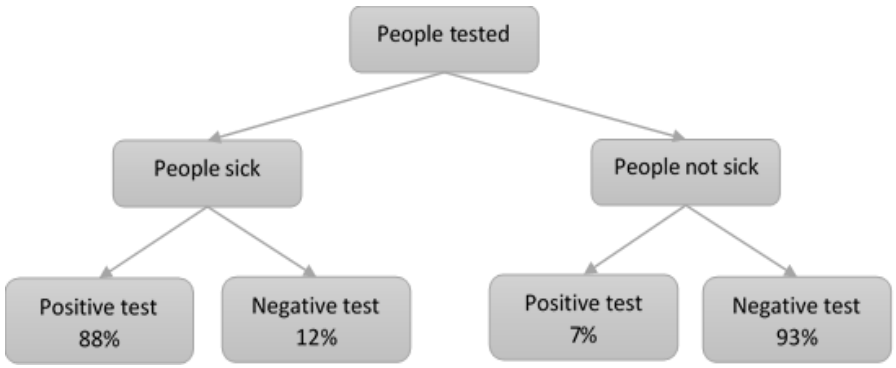
- $2 bet:
  - Win $2 with probability 0.5, or $0 with probability 0.5.
  - A $2 bill and a coin: flip the coin to see if you get the $2.
- $1 bet:
  - Win $1 with probability 0.99 or $1,000,000 with probability 0.00000001.
  - A $1 bill and a lottery ticket: if the ticket is a winner, you get $1,000,000, else you keep the $1 bill.
- $3 bet:
  - Win $3 with probability of 0.999999 or lose $1,999,997 with probability 0.000001.
  - A box containing a million quarters; all are NY quarters, except for a NJ one; pick one out from the box; if NY quarter, you win $3; if NJ quarter, you owe almost $2M.

Real-life Risk Analysis

- Case 1:
  - A sure gain of $100.
  - A 50% chance to win $200 and a 50% chance to win $0.
- Case 2:
  - A sure loss of $100.
  - A 50% chance to lose $200 and a 50% chance to lose $0.

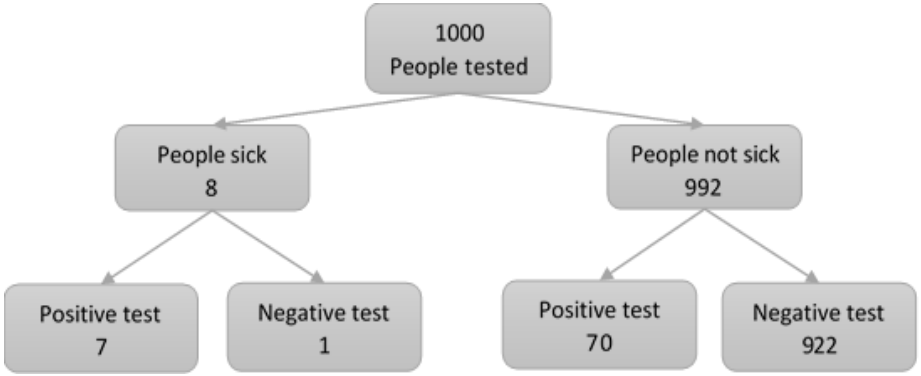
Real-life Risk Analysis

- How people think:



Most people with positive tests are sick

Actual numbers:



7 out of 77 (9%) positive tests are sick

- $$P(I|T) = \frac{P(T|I)\times P(I)}{P(T|I)\times P(I)+P(T|\bar{I})\times P(\bar{I})} = \frac{0.88\times 0.008}{0.88\times 0.008+0.07\times 0.992} = 9\%$$

Real-life Risk Analysis

- Person X receives news of a positive screen test and wonders what is the probability of being sick. The following information is available:
  - Probability that a person tested is actually sick is 0.8%.
  - If a person is sick, probability for a positive test is 88%.
  - If a person is not sick, probability for a positive test is 7%.

  - Most people focus on the second statement because it implies causality.
  - First statements is ignored as being statistics.
  - Third statement is ignored as not being about the category of interest.
  - Then, conclude that most people with positive tests are sick.

Extreme and Rare Events

- Extreme events:
  - Phenomena that have relatively extreme high (or low) degree of magnitude.
- Rare events:
  - Phenomena that have relatively very low frequency of occurrence.
- Extreme and rare events:
  - Phenomena that have relatively very low frequency of occurrence and at the same time have relatively extreme high (or low) degree of magnitude.
  - Relative difference between the magnitudes and frequency of occurrence of these events is the primary reason for which they are difficult to analyze and manage.

Risk of Extreme (and Rare) Events

- Impact of highly improbable.
  - Understanding produced by assessing signals after an event is usually not general enough to yield insight, misses causal understanding, and often lacks meaningful impact on decisions.
  - Improbable events that result in the greatest impact (e.g., the unimagined terrorist attack on passenger planes; the implausible combination of hurricane and a failure of critical infrastructure).
- Probabilities and associated consequences defy any common expected value representation of risk.

Risk of Extreme (and Rare) Events

- Three attributes of an extreme event.
  - It is an outlier, as it lies outside the realm of regular expectation; nothing in the past can convincingly point to its possibility.
  - It carries an extreme impact.
  - In spite of its outlier status, human nature makes us come up with explanations for its occurrence after the fact, making it explainable and predictable.
- Besides mathematics and engineering, includes social and behavioral sciences, natural sciences, and economics sciences.
- Defies the reliance on the expected value of risk.

Fallacy of Expected Value

- Imagine what life would be like if:
  - Highways were constructed to accommodate the average traffic load of vehicles of average weight.
  - Mass transit systems were designed to move only the average number of passengers (i.e., total number of passengers per 24 hours) during each hour of the day.
  - Bridges, homes, and industrial/commercial systems were constructed to withstand the average wind or earthquake.
  - Telephone lines and switchboards were sufficient in number to accommodate only the average number of calls per hour.
  - Electric utility calculated the year-round average electrical demand during all hours of day and all seasons of year.
  - Emergency services provided only the average number of personnel and facilities during all hours of the day and all seasons of the year.
  - Space programs provided emergency procedures for only the average type of failure.

Fallacy of Expected Value

- Two events:
  - Event E_1: 2 PPB pollutant.
  - Event E_2 = 20,000 PPB pollutant.
- Probabilities of occurrence:
  - $p_1 = 0.1$ and $p_2 = 0.00001$.
- Expected values:
  - EV (E_1) = 0.1*2 = 0.2
  - EV (E_2) = 0.00001*20,000 = 0.2
- Analysis:
  - Low likelihood of a disastrous contamination with 20,000 PPB pollutant cannot be equivalent to the contamination at a low concentration of 2 PPB, even with a high likelihood.

Fallacy of Expected Value

- A low probability, but high impact event cannot be treated in the same way as other events, just because their expected values are the same.
  - A catastrophic dam failure, which causes flooding of 10^6 acres, with the associated damage to human life and the environment, but with a probability of occurrence of 10^{-6}.
  - Minor flooding of 10^2 acres of land with a probability of happening of 10^{-2}.
- Two questions:
  - How safe is safe enough?
  - What is an acceptable level of risk?
  - Expected value of risk does not accentuate the catastrophic events and their consequences, thus misrepresenting what would be perceived as an unacceptable risk.

Fallacy of Expected Value

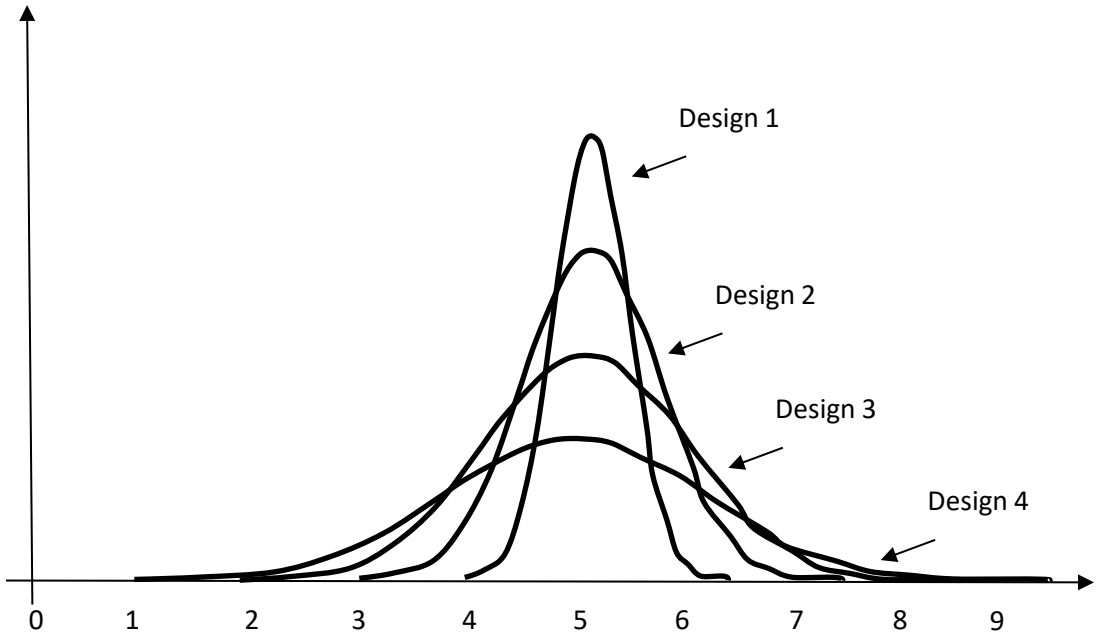
- Four different designs are considered.

Design	Cost ($)	Failure rate mean	Failure rate standard deviation
1	100,000	5	1
2	80,000	5	2
3	60,000	5	3
4	40,000	5	4

  - Based on the expected value, design 4 is selected.

Fallacy of Expected Value

- Four different designs are considered.



  - Considering the variances, which provide an indication of extreme failures, design 4 might not be the best.
  - More in depth analysis is needed.

Solutions

- Multi-criteria decision and risk analysis (MCDA).
- Utility theory.
- Partitioned multi-objective risk method (PMRM).
- Extreme value analysis.
  - Focus interest is in the parametric model of the distribution of the maxima and the exceedances.
  - Involves the tails of distributions.
  - Probability of exceedance of a random variable x is the probability that it will be exceeded.

Decision Involving Multiple Criteria

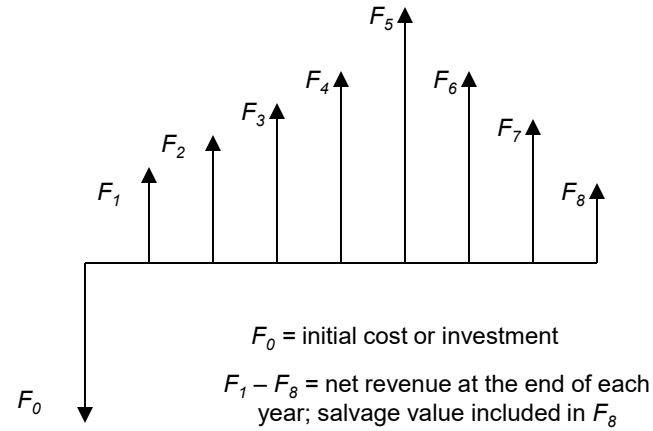
- Decisions must be made based on multiple criteria that jointly influence the relative desirability of alternatives under consideration.
- Select criteria to be considered.
  - Criteria need be independent of each other.
  - Number of criteria selected should be kept at minimum.
- Differences are the basis of decision.
  - Criteria that are identical are canceled out.

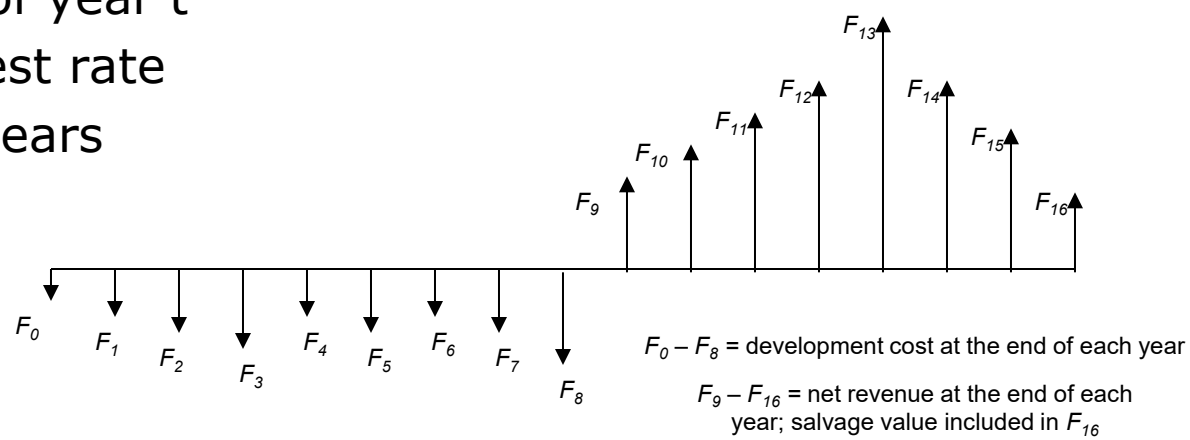
Decision Evaluation Theory

- Choose among alternatives during system design, and/or optimize systems already in operation.
  - Money flow modeling.
  - Economic optimization modeling.

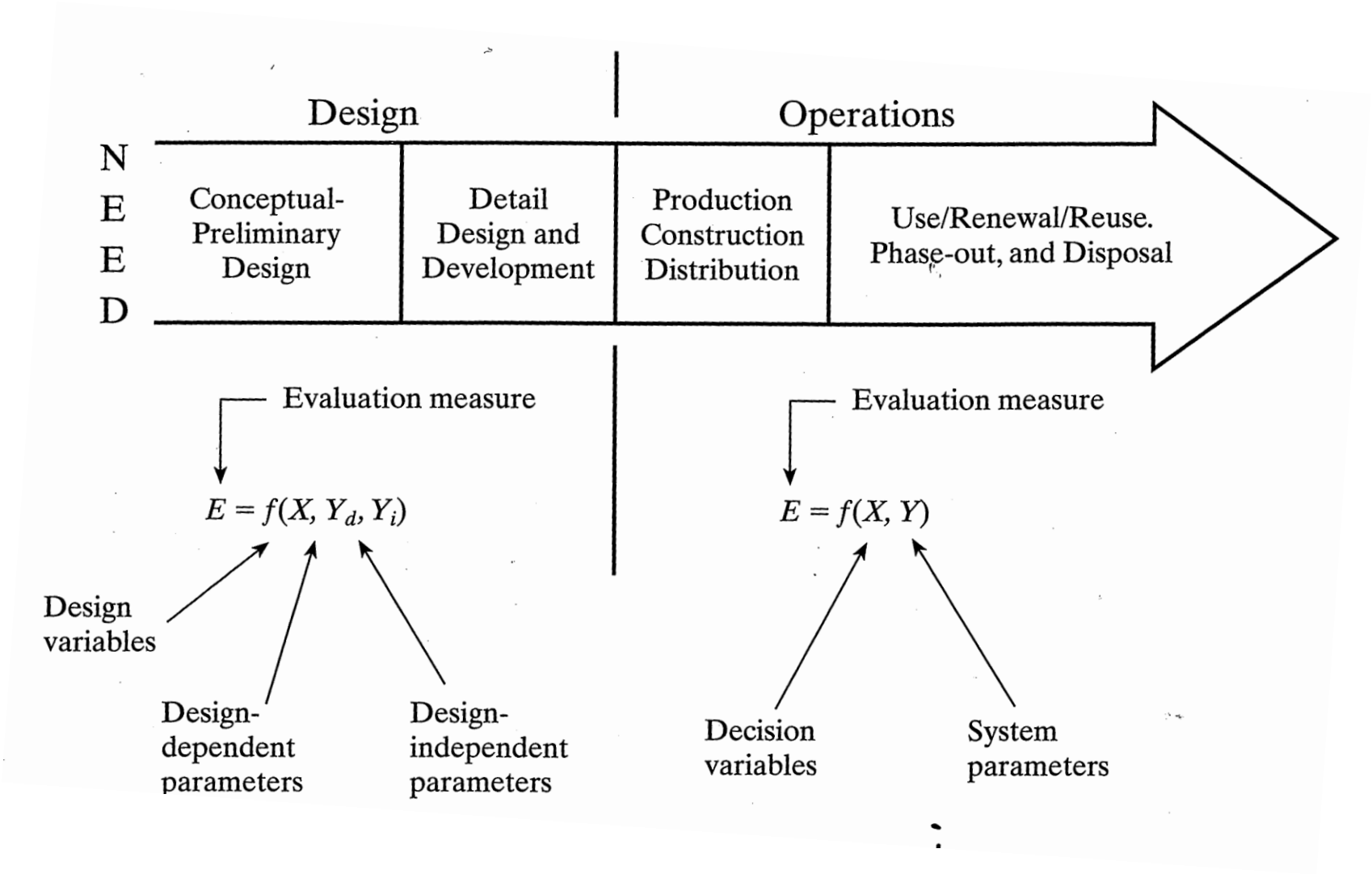
Evaluation by Money Flow Modeling

- Economic equivalence is expressed as present equivalent, annual equivalent, future equivalent (also, internal rate of return, payback period)
- *PE, AE,* or *FE* = $f(F_t, i, n)$
  - t = 0, 1, 2,…, n
  - F_t = positive or negative money flow at the end of year t
  - i = annual interest rate
  - n = number of years





Evaluation by Economic Optimization Modeling

- Mathematical model linking an evaluation measure, E, with controllable decision variables, X, and system parameters, Y, which cannot be controlled by the decision maker.
- Generally: $E = f(X, Y)$
- System design: $E = f(X, Y_d, Y_i)$
- Design-dependent parameters (Y_d)
  - Factors under the control that are impacted by specialty disciplines during development process.
  - Reliability, maintainability, throughput.
- Design-independent parameters (Y_i)
  - Factors beyond the control, which impact the effectiveness of all candidate systems or design alternatives.
  - Labor rates, material cost, interest factors.

Evaluation by Economic Optimization Modeling